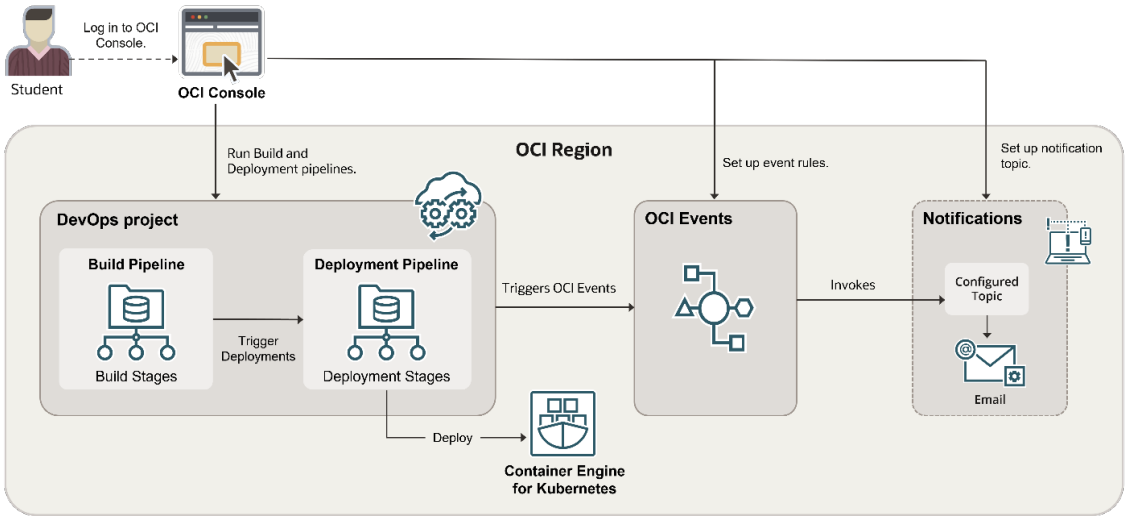
Lab 17. Define Rules that Trigger a Specific Action When a DevOps Event Occurs

Oracle Cloud Infrastructure (OCI) Events enables the creation of automations based on resource state changes across the tenancy. Use Events to allow your development teams to react automatically when a resource changes state.

Events are structured messages indicating changes in resources.

The use of events goes through the creation of *rules*. The rules include a filter that you define to indicate events produced by resources in your tenancy.



In this lab, you'll:

a. Configure a notification
b. Create an event rule
c. Validate the event rule by running a build

Prerequisites

- You must complete the following labs before you perform tasks for this practice:
  - *Create Docker image for a web application using Dockerfile*.
  - *Create and work with OCIR repository.*
  - *Deploy a load-balanced Web application on an OKE cluster using Kubectl.*
  - *Work with code repositories in OCI DevOps project.*
  - *Create and set up artifacts and environments in DevOps project.*
  - *Deploy a Web App with a CI/CD pipeline to an OKE cluster using OCI DevOps*

Assumptions

- This lab assumes you're working in the Ashburn region. `(iad)`.
- You will replace the *`<userID>`* placeholder with your firstname.

Configure a Notification

Events Rules specify an action to trigger when the filter finds a matching event. The Action can be the Notifications service to notify based on the rule conditions defined.

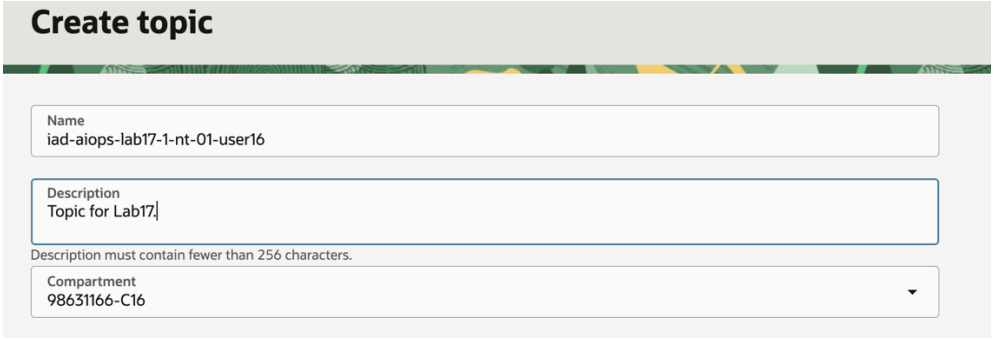
To configure an Events Rule, you must first create a Notifications Topic and Subscription so that the rule condition has a way to notify the relevant parties.

1. Open the navigation menu and select **Developer Services**. Under **Application Integration**, select **Notifications**.
2. Select your *`<assigned compartment>`* from List scope on the left menu.
3. Click **Create Topic** and enter the following values in the form:
   - **Name** `iad-aiops-lab17-1-nt-01-<userID>`

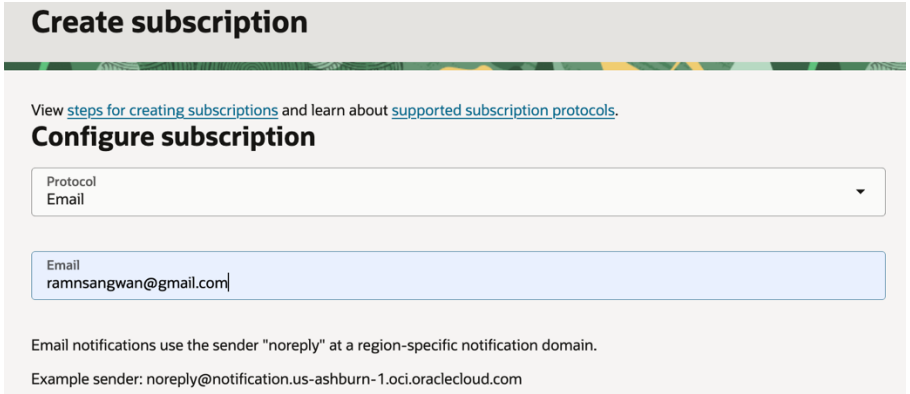
     For Example, iad-aiops-lab17-1-nt-01-user16

   - **Description** `Topic for Lab17.`

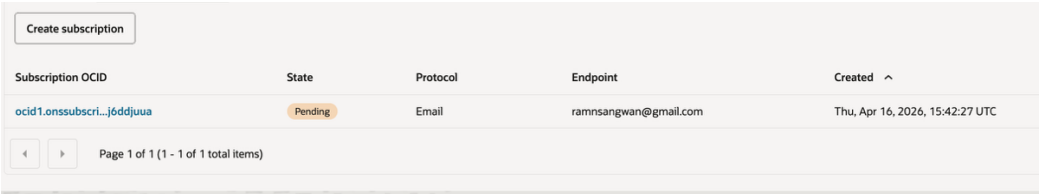
   Topic name must be unique across the tenancy; validation is case-sensitive.



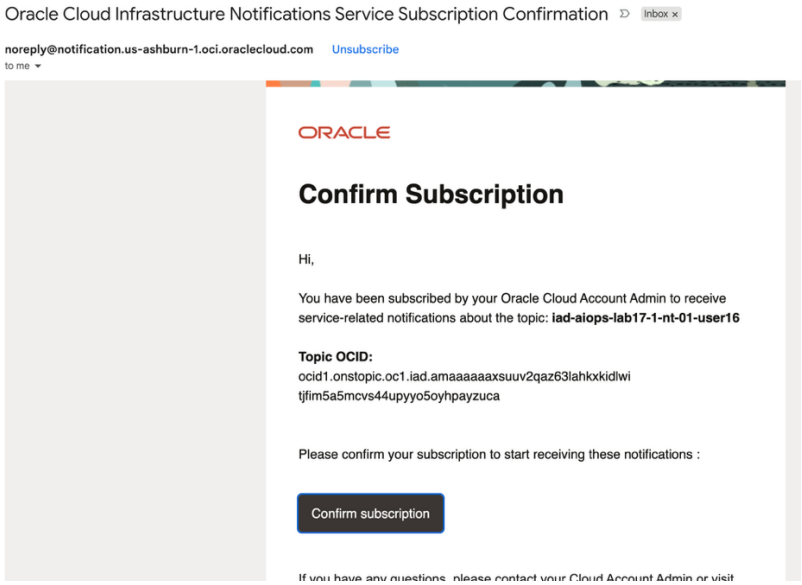
4. Click **Create**.
5. Once the topic changes state to **Active**, click the topic to view the details.
6. Click on **Subscriptions** tab and then click **Create Subscription** and enter the following values in the form:
   - **Protocol**: Select **Email**.
   - **Email:** Enter your email address.



7. Click **Create**.
8. Click the subscription that you just created. The Subscription Information will be displayed with the status as **Pending**.

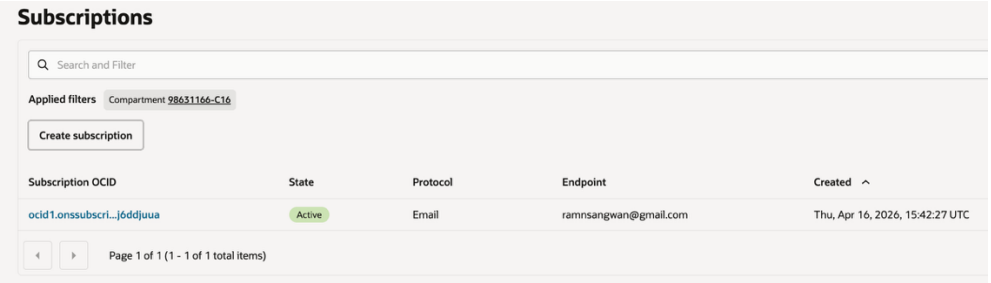


9. Check the verification email received on the email account you specified. Click the **Confirm subscription verification link**. A pop-up browser window will tell you that the subscription has been confirmed.



If the email does not arrive in the inbox, look for it in the Promotions category or the spam box.

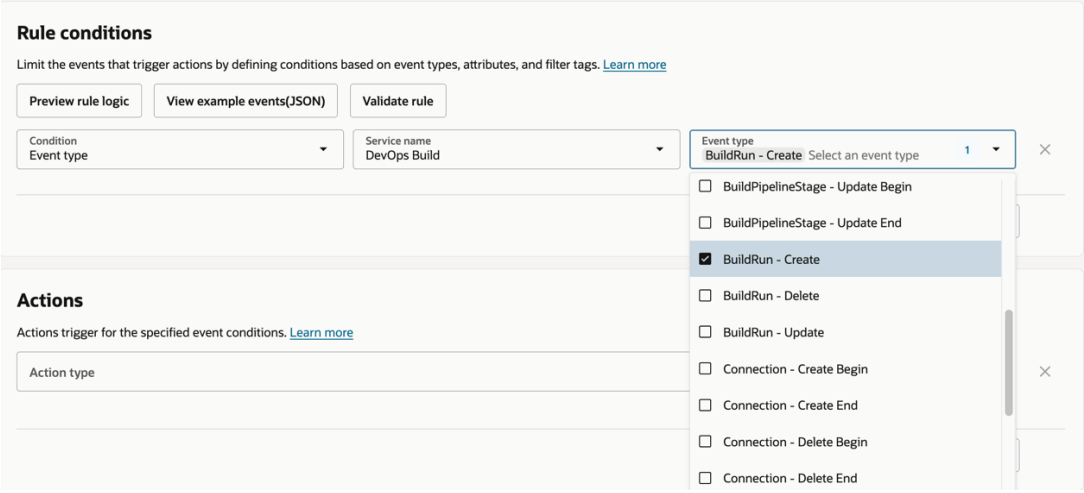
10. Navigate back to the **Subscriptions** page and verify that the subscription status has changed to **Active**. You may need to refresh your browser if the status is not updated.



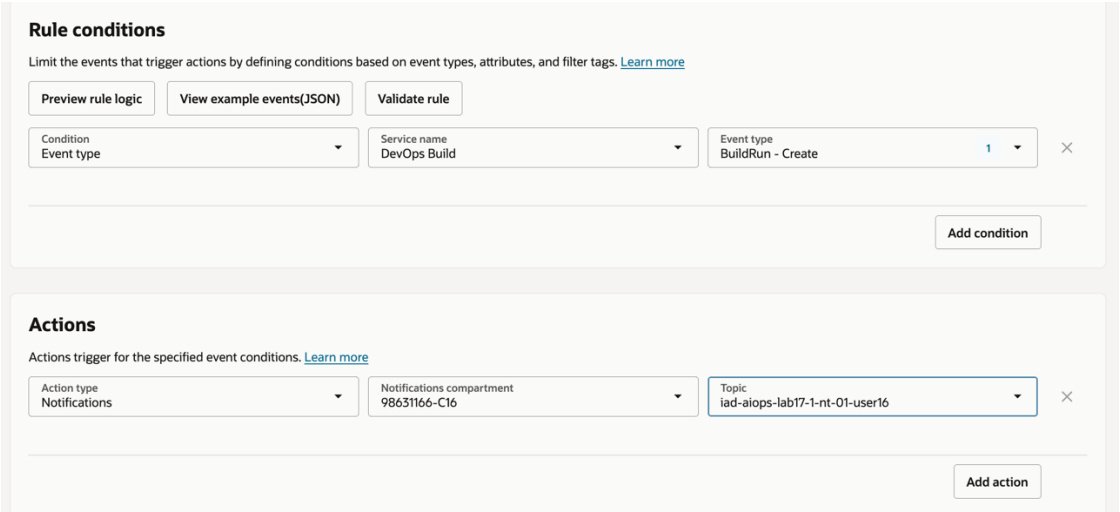
Create an Event Rule

You will create Events Rule with few conditions and an action to send a notification.

1. Open the navigation menu and select **Observability & Management**. Under **Event Service**, click **Rules**.
2. Select your `<assigned compartment>` from Applied filters.
3. Click **Create Rule** and enter the following values in the form:
   a. **Display Name:** `IAD-AIOPS-LAB17-1-RLE-01`
   b. **Description:** `Notify on Build Runs and Deployments.`
   c. Under the **Rule Conditions** section, choose Condition as **Event Type** and Service Name as **DevOps Build.**
   d. Click within Event Type field and select **BuildRun – Create**



   e. Under **Actions**, select the following:
      - **Action Type**: Notifications
      - **Notification Compartment**: Select your `<assigned compartment>`.
      - **Topic**: iad-aiops-lab17-1-nt-01-<userID>



4. Click **Create Rule**.
5. Click **Rules** on top; verify the State is shown as **Active.**

Validate Event Rule by Running a Build

Execute a manual build run from DevOps Project. After the build run, an email will be triggered with event details.

1. Open the navigation menu and select **Developer Services**. Under **DevOps**, select **Projects**.
2. Select your `<assigned compartment>` from List scope on the left menu.
3. Open the DevOps project **IAD-DOP-LAB05-1-DP-01-<*userID*>**.
4. Select **Build Pipelines** on the left menu and click the **IAD-DOP-LAB11-1-BPL-01** pipeline**.**
5. Verify the three stages are available: Build WebApp, Push WebApp Artifacts, and Trigger OKE Deployment.
6. Click **Start manual run** on the top right. Enter the Build run name as **IAD-DOP-LAB17-1-BRUN-01**.
7. Click **Start manual run.** A Build Run is created with name IAD-DOP-LAB17-1-BRUN-01.
8. You will receive an email with the subject line:

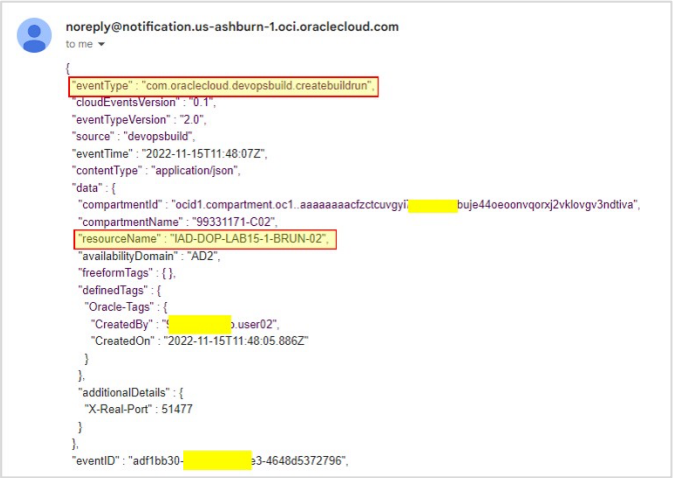
   Event - com.oraclecloud.devopsbuild.createbuildrun along with Event details in the body in JSON format, including: resourceName as IAD-DOP-LAB17-1-BRUN-01.

noreply@notification.us-ashburn-1.oci.oraclecloud.com
to me


```
{
 "eventType" : "com.oraclecloud.devopsbuild.createbuildrun",
 "cloudEventsVersion" : "0.1",
 "eventTypeVersion" : "2.0",
 "source" : "devopsbuild",
 "eventTime" : "2026-04-16T15:52:45Z",
 "contentType" : "application/json",
 "data" : {
  "compartmentId" : "ocid1.compartment.oc1..aaaaaaaawjohov7fktaf7b7hhqkh7xtjjn2l44pnwcy42dhvqjziyllkxokq",
  "compartmentName" : "98631166-C16",
  "resourceName" : "IAD-DOP-LAB17-1-BRUN-01",
  "availabilityDomain" : "AD1",
  "freeformTags" : { },
  "definedTags" : { },
  "additionalDetails" : {
   "X-Real-Port" : 65406
  }
 },
 "eventID" : "6da31bc6-16af-43e3-8333-cc7a11e7d20d",
```


9. Open the navigation menu, select **Observability & Management**. Under **Events**, click **Rules**.
10. Select your `<assigned compartment>` from List scope on the left menu.
11. Click **IAD-AIOPS-LAB17-1-RLE-01** and click **Edit Rule**.
12. Under **Rule Conditions**, click **+ Another Condition** to add a second condition as follows:
    - **Condition**: Event Type
    - **Service Name**: Devops Deploy
    - Click within Event Type field and select: **DeployStage – Create Begin**, and **Deployment – Create**.
13. Click **Save changes.** After updating the event rule, execute another manual build run to receive email notification according to the updated event rules.
14. Open the **DevOps project IAD-DOP-LAB05-1-DP-01-<*userID*>**.
15. Select **Build Pipelines** on the left menu and click the **IAD-DOP-LAB11-1-BPL-01** pipeline**.**
16. Verify the three stages are available: Build WebApp, Push WebApp Artifacts, and Trigger OKE Deployment.
17. Click **Start manual run** on top right. Enter the Build run name as **IAD-DOP-LAB17-1-BRUN-02**.
18. Click **Start manual run.** A Build Run is created with name IAD-DOP-LAB17-1-BRUN-02.
19. You will receive an email with the subject line

    Event - com.oraclecloud.devopsbuild.createbuildrun along with Event details in the body in JSON format, including: resourceName as IAD-DOP-LAB17-1-BRUN-02.



20. You will receive another email with the subject line **Event - com.oraclecloud.devopsbuild.createdeployment** along with Event details in the body in JSON format.

Congratulations! You have successfully configured event rules to trigger email notifications for specific actions.

Purge Instructions

Purge Instructions for Event Rule

1. Open the navigation menu and select **Observability & Management**. Under **Events**, click

Rules.

2. Select your *<assigned compartment>* from List scope on the left menu.
3. For the rule **IAD-AIOPS-LAB17-1-RLE-01**, click the three dots on the right to open the Actions menu and select **Delete**
4. Type **DELETE** to confirm, click **Delete.**

Purge Instructions for Subscription

1. Open the navigation menu and select **Developer Services**. Under **Application Integration**, click **Notifications**.
2. Select your *<assigned compartment>* from List scope on the left menu.
3. Click the topic **IAD-DOP-LAB17-1-TOP-01**.
4. For the subscription, click the three dots on the right to open the Actions menu and select **Delete**.
5. Click **Delete Subscription** to confirm.

Purge Instructions for Topic

1. Open the navigation menu and select **Developer Services**. Under Application Integration, click **Notifications**.
2. Select your *<assigned compartment>* from List scope on the left menu.
3. For the topic **IAD-DOP-LAB17-1-TOP-01**, click the three dots on the right to open the Actions menu and select **Delete**.
4. Click **Delete Topic** to confirm.